

Examination Scheme for Semester-VII

CA – Continuous Assessment

EoSE – End of Semester Examination

Regular Students –

Type of Examination	Course Code and Nomenclature	Scheme of Examination	Duration of Examination	Maximum Marks	Minimum Marks
Theory	UG0804 -BTH-87T-751 RECOMBINANT DNA TECHNOLOGY	CA	01 Hr	20 Marks	08 Marks
		EoSE	03 Hrs	80 Marks	32 Marks
Practical	UG0804 -BTH-87P-752 RECOMBINANT DNA TECHNOLOGY-PRACTICAL	CA	1 Hr	10 Marks	04 Marks
		EoSE	04 Hrs	40 Marks	16 Marks
Theory	UG0804 -BTH-87T-753 SCIENTIFIC WRITING AND PRESENTATION	CA	01 Hr	20 Marks	08 Marks
		EoSE	03 Hrs	80 Marks	32 Marks
Practical	UG0804 -BTH-87P- 754 SCIENTIFIC WRITING AND PRESENTATION-PRACTICAL	CA	1 Hr	10 Marks	04 Marks
		EoSE	04 Hrs	40 Marks	16 Marks
Theory	UG0804 -BTH-87T-755 FOOD AND DAIRY TECHNOLOGY	CA	01 Hr	20 Marks	08 Marks
		EoSE	03 Hrs	80 Marks	32 Marks
Practical	UG0804 -BTH-87P-756 FOOD AND DAIRY TECHNOLOGY- PRACTICAL	CA	1 Hr	10 Marks	04 Marks
		EoSE	04 Hrs	40 Marks	16 Marks

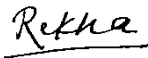
The theory question paper will consist of **two** parts, **A & B**.

PART-A: 20 Marks

Part A will be compulsory, having 10 very short answer-type questions (with a limit of 20 words) of two marks each.

PART-B: 60 Marks

Part B of the question paper shall be divided into four units comprising question numbers 2-5. There will be one question from each unit with an internal choice. Each question will carry 15 marks.

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UG0804 – Three/Four Year Bachelor of Science (Bio-Technology)								
				Bio-Technology 4 th YEAR SEM-VII			Credits	
#	L e v e l	S e m e s t e r	Type	Title	L	T	P	Total
1.	8	VII	MJR	UG0804 -BTH-87T-751 RECOMBINANT DNA TECHNOLOGY	4	0	0	4
2.	8	VII	MJR	UG0804 -BTH-87P-752 RECOMBINANT DNA TECHNOLOGY-PRACTICAL	0	0	2	2
3.	8	VII	MJR	UG0804 -BTH-87T-753 SCIENTIFIC WRITING AND PRESENTATION	4	0	0	4
4.	8	VII	MJR	UG0804 -BTH-87P- 754 SCIENTIFIC WRITING AND PRESENTATION-PRACTICAL	0	0	2	2
5.	8	VII	MJR	UG0804 -BTH-87T-755 FOOD AND DAIRY TECHNOLOGY	4	0	0	4
6.	8	VII	MJR	UG0804 -BTH-87P-756 FOOD AND DAIRY TECHNOLOGY-PRACTICAL	0	0	2	2

BTH-87T-751 RECOMBINANT DNA TECHNOLOGY

Semester	Code of the Course	Title of the Course/Paper			NHEQ F Level	Credits
VII	BTH-87T-751	RECOMBINANT DNA TECHNOLOGY			8	4
Level of Course	Type of the Course	Credit Distribution			Offered to NC Student	Course Delivery Method
		Theory	Practical	Total		
Advanced	Major	4	2	6	NO	60 lectures with diagrammatic and informative assessments during lecture hours

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List of Programme Codes in which Offered as Minor Discipline	-----
Prerequisites	Knowledge of the Intermediate level
Objectives	➤ To familiarize students with the fundamental concepts and tools of genetic engineering and recombinant DNA technology. ➤ To impart knowledge about enzymes, vectors, and techniques used in gene cloning and DNA manipulation. ➤ To understand applications of recombinant DNA technology in agriculture, healthcare, and industry. ➤ To develop skills in gene transfer methods, DNA analysis, and interpretation of results.

Detailed Syllabus
BTH-87T-751 RECOMBINANT DNA TECHNOLOGY

Unit-I

Introduction and Tools of Recombinant DNA Technology: Historical perspective and milestones in recombinant DNA technology. Enzymes used in rDNA technology: Restriction endonucleases, ligases, DNA polymerases, reverse transcriptase, polynucleotide kinase, and alkaline phosphatase. Methods for the isolation of nucleic acid. Gel electrophoresis and blotting techniques (Southern, Northern, Western).

15 Lectures

Unit-II

Vectors and Gene Cloning Strategies: Cloning vectors: Plasmid vectors, bacteriophage vectors (lambda, M13), cosmids, phagemids, BACs and YACs. Expression vectors and shuttle vectors. Construction of recombinant DNA molecules (procedure of cloning a gene of interest). Methods of introducing DNA into host cells: Transformation, transfection, electroporation, microinjection, biolistics. Selection and screening of recombinants: Blue-white screening, colony hybridization, and PCR-based screening.

15 Lectures

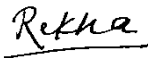
Unit-III

PCR, DNA Sequencing, and Analysis: Polymerase Chain Reaction (PCR): Principles, types (RT-PCR, multiplex PCR). DNA sequencing methods: Sanger sequencing, Next-Generation Sequencing (NGS) overview. DNA fingerprinting and its applications. Restriction Fragment Length Polymorphism (RFLP), RAPD, AFLP, and microsatellite analysis. Bioinformatics tools for sequence analysis and interpretation.

15 Lectures

Unit-IV

Applications of Recombinant DNA Technology: Production of recombinant proteins: Insulin,

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growth hormones, vaccines. Transgenic plants and animals: Methods, advantages, and examples (Bt cotton, golden rice). Gene therapy: Types, methods, and applications. Ethical, safety, and regulatory issues in recombinant DNA technology. Intellectual Property Rights (IPR).

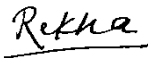
15 Lectures

BTH-87P-752 RECOMBINANT DNA TECHNOLOGY –PRACTICAL

1. Isolation of genomic DNA from human cheek cells/ plant tissue.
2. Isolation of plasmid from a prokaryotic cell (alkaline lysis method).
3. Agarose gel electrophoresis for DNA analysis.
4. Assessment of DNA purity and quantity using spectrophotometry.
5. Culture of *E. coli* and yeast, including streaking and subculture.
6. Preparation of Competent Cells (CaCl₂ method)
7. PCR amplification of a target DNA sequence
8. Restriction digestion of DNA
9. DNA ligation
10. Transformation of bacteria and selection of transformations
11. Screening of transformations
12. Blue/white screening
13. Protein profiling by SDS-PAGE
14. Demonstration of cDNA Library Preparation (Chart/Video)
15. Southern Blotting (Demonstration or Chart)
16. DNA fingerprinting demonstration using gel electrophoresis.
17. Bioinformatics tools: BLAST, sequence alignment, and ORF prediction.
18. Visit to a Plant Tissue Culture/Transgenic Facility (Virtual/Physical).
19. Visit to a molecular biology laboratory or biotechnology industry for exposure.
20. Any other exercise based on the theory paper.

Suggested Books & Text

1. A textbook of biotechnology – R. C. Dubey
2. Biotechnology Expanding Horizons– B. D. Singh
3. Gene Cloning and DNA Analysis- T.A. Brown
4. Concepts of Genetics – Kluggs and Cummings
5. Molecular Biotechnology- Bernard R. Glick
6. Textbook of biotechnology –H.K. Das
7. Introduction to Plant Biotechnology- H. S. Chawala
8. Principles of Gene Manipulation and Genomics - Primrose and Twyman
9. Genetic Engineering Principles & Practice – Sandhya Mitra
10. From genes to clones- Winacker
11. Gene Cloning- T. A. Brown
12. From genes to genome: concepts and applications of DNA technology- Dale and Schantze
13. Gene Cloning: Principles and Applications- Lodge, Lund and Minchin
14. Gene Biotechnology- S. N. Jogdanand

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15. Biotechnology- U. Satyanarayan
16. Genetic Engineering – Smita Rastogi & Neelam Pathak
17. An Introduction to Genetic Engineering. Nicholl, D.S.T.
18. Molecular Biology and Genetic Engineering. Gupta, P. K.
19. Molecular Biotechnology: Principles and Applications of Recombinant DNA. Glick, B. R., and Pasternak, J. J.
20. Molecular Cloning: A Laboratory Manual. Sambrook, J., and Russell, D.W. (Cold Spring Harbor).
21. Recombinant DNA: Genes and Genomes. Watson, J.D., et al.

Online Resources: NCBI, EMBL, KEGG Pathway Databases for Practical Bioinformatics Exposure.

Course Learning Outcomes:

On successful completion of this course, students will be able to:

1. Explain the basic principles of recombinant DNA technology and describe the roles of enzymes, vectors, and molecular tools used in gene cloning.
2. Recall key discoveries, gene transfer methods, and types of cloning and expression vectors used in genetic engineering.
3. Perform essential techniques such as plasmid isolation, PCR, restriction digestion, DNA ligation, transformation, and gel electrophoresis.
4. Analyze DNA samples using PCR-based methods, electrophoresis, and DNA fingerprinting for gene detection and variation studies.
5. Describe the production of recombinant proteins (e.g., insulin, vaccines) and the development of transgenic organisms for use in agriculture and medicine.
6. Demonstrate skills in screening of recombinants using blue-white selection, colony hybridization, and bioinformatics tools such as BLAST and sequence alignment.
7. Interpret sequencing results and compare traditional (Sanger) and advanced (NGS) DNA sequencing techniques.
8. Discuss applications, ethical concerns, biosafety guidelines, and IPR or ethical issues associated with recombinant DNA technology and genetic engineering.

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BTH-87T-753 Scientific Writing and Presentation

Semester	Code of the Course	Title of the Course/Paper			NHEQ F Level	Credits
VII	BTH-87T-753	Scientific Writing and Presentation			8	4
Level of Course	Type of the Course	Credit Distribution			Offered to NC Student	Course Delivery Method
		Theory	Practical	Total		
Advanced	Major	4	2	6	NO	60 lectures with diagrammatic and informative assessments during lecture hours
List of Programme Codes in which Offered as Minor Discipline		-----				
Prerequisites		Knowledge of the Intermediate level				
Course Objectives		➤ To enable students to compose clear, error-free scientific texts, use Word and Excel for editing and data handling, and draft formal correspondence (letters, memos, agendas). ➤ To enhance students' skills to plan and write structured technical reports, produce coherent project/internship reports, and create effective emails, brochures, and leaflets. ➤ To enable students to formulate research questions, search scholarly sources, draft full research proposals using proper citation styles, and practice ethical writing (avoiding plagiarism). ➤ To enable students to design engaging visuals and posters, use tools like PowerPoint, LaTeX Beamer, Adobe, and CorelDRAW, and deliver confident oral presentations with effective audience engagement and Q&A handling.				

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Detailed Syllabus
BTH-87T-753 Scientific Writing and Presentation

UNIT-I

Scientific Writing & Digital Tools: Scientific Writing Fundamentals: Introduction to clear, concise scientific English; structuring paragraphs and writing short assignments; efficient note-taking strategies. **Common Writing Pitfalls:** Identifying and avoiding errors—grammar, punctuation, syntax, clarity; self-editing and revision techniques. **Digital Editing & Data Handling:** Mastering editing tools in Microsoft Word and Office; accurate data entry and spreadsheet management using Excel (basic formulas, formatting, tables). **Professional Correspondence:** Crafting formal and informal letters; formatting memos, agendas, meeting minutes, notices, and circulars using standard formats.

15 Lectures

UNIT-II

Report Writing & Professional Documents: Report Writing Essentials: Purpose, scope, and structural principles of technical reports. **Project & Internship Reports:** Writing structured, coherent project reports and summer internship reports for academic and industry settings. **Professional Communication Materials:** Designing clear and effective emails, brochures, and leaflets; adapting style and tone for diverse professional audiences.

15 Lectures

UNIT-III

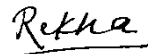
Research Proposals & Ethical Writing: Research Design Skills: Identifying research problems/questions/hypotheses; conducting literature searches; organizing information effectively; exploring digital scholarly tools—e-resources, e-journals, INFLIBNET, Shodhganga. **Research Proposal Drafting:** Writing structured proposals: Title & Abstract, Introduction/Background, Literature Review, Objectives, Methodology, Expected Outcomes, **Research Ethics & Integrity:** plagiarism avoidance, proper attribution, and responsible scientific communication with ethical rigor.

15 Lectures

UNIT-IV

Visual & Oral Presentation Skills: Planning and crafting presentations; principles for effective slide layouts and visual information packaging. **Oral Presentation Techniques:** Structuring and delivering an engaging talk; managing nerves and employing nonverbal communication; engaging the audience; handling Q&A effectively. **Poster & Public Display Skills:** Designing and presenting scientific posters and handouts for conferences and academic events with clarity and professionalism.

15 Lectures

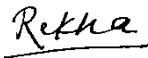
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BTH-87P-754 SCIENTIFIC WRITING AND PRESENTATION–PRACTICAL

1. Write a 150–200-word summary on a given scientific topic with clear, concise English.
2. Listen to a 5-minute audio/video on a scientific topic and prepare effective bullet-point notes.
3. Edit a provided draft (with intentional errors) using track changes in MS Word.
4. Practicals on using spelling and grammar checks, comments, track changes, and styles in MS Word.
5. Enter a small dataset (5–6 columns, 10 rows), perform formatting, use basic formulas (SUM, AVERAGE), and create a simple table.
6. Draft a formal leave application, an informal email to a friend, and a thank-you email using clear formatting.
7. Prepare a memo and a notice for a departmental meeting using standard formats.
8. Prepare the structure of a technical report (Title, Abstract, Introduction, Methods, Results, Conclusion).
9. Write a one-page project report on a lab experiment conducted during the semester.
10. Prepare a template for a summer internship report with proper headings and subheadings.
11. Write a professional email requesting a lab visit appointment with appropriate tone and structure.
12. Design a one-page brochure for a lab workshop or departmental event using MS Word/Canva.
13. Search for 2–3 articles using INFLIBNET/Shodhganga/Google Scholar and prepare a reference note.
14. Draft a one-page mini-research proposal with Title, Objectives, Methodology, and Expected Outcomes.
15. Check a provided text for plagiarism using a free tool and write steps to avoid plagiarism in writing.
16. Prepare a 5-slide scientific presentation using effective slide layouts and concise bullet points.
17. Design an academic poster on a given topic using PowerPoint/Canva.
18. Create a bar graph and a table in Excel and insert them into a Word document.
19. Design a simple infographic explaining a scientific process using Canva.
20. Deliver a 3-minute oral presentation on a chosen scientific topic.

Suggested Books & Text

1. Steven, Michael. *How to Be a Better Problem Solver*. Kogan Page, New Delhi, 1999
2. Raman, Prakash. *Business Communication*. Oxford University Press.
3. Krishnaswamy, N. *Creative English for Communication*. Macmillan.
4. Ramaswamy, S. *Textbook of Business Communication*. Macmillan.

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5. Withrow, Jean. *Effective Writing*. Cambridge University Press.
6. Coe, Norman, Robin Rycroft, and Pauline Ernest. *Writing Skills: A Problem-Solving Approach for Upper-Intermediate and More Advanced Students*. Cambridge University Press.
7. Mitra, Barun K. *Effective Technical Communication: A Guide for Scientists and Engineers*. Oxford University Press India, 2006.
8. Raman, Meenakshi, and Sangeeta Sharma. *Technical Communication: Principles and Practice*. 4th ed., Oxford University Press India, 2022.
9. Raman, Meenakshi, and Sangeeta Sharma. *Professional Communication*. 3rd ed., Oxford University Press India, 2017.
10. Kalpana, S., and K. Kanimozhi. *Scientific Writing Handbook*. CBS Publishers, 2025.
Parija, Subhash C., and Peush Sahni. *Reporting and Publishing Research in the Biomedical Sciences*. Springer, 2017.
11. Singh, Kalpana, Monisha Banerjee, and Jyotika Rajawat. *A Concise Guide on Report Writing and Presentation Skills*. IIP Publishers, 2023.
12. Singh, Kalpana. *A Concise Guide to Scientific Writing & Communication*. IIP Publishers, 2024.
13. Sholapurkar, Amar A. *Publish and Flourish: A Practical Guide for Effective Scientific Writing*. Jaypee Brothers, 2021.
14. Becker, Lucinda, and Pam Denicolo. *Presenting Your Research: Conferences, Symposia, Poster Presentations and Beyond*. SAGE Publications, 2013.
15. Hull, Charles L. H. *How to Give a Great Talk*. arXiv, 2020.

Course Learning Outcomes:

At the end of this unit, students will be able to:

1. Compose clear, grammatically correct English tailored to scientific content, including well-structured paragraphs and effective note-taking.
2. Identify and correct common writing errors, revising drafts toward concision and precision.
3. Draft professional correspondence (letters, memos, agendas, minutes, notices, circulars) using appropriate formats and tone.
4. Plan and structure formal scientific reports, understanding their purpose and audience.
5. Write detailed project and internship reports, organizing content logically and formatting consistently.
6. Create effective emails, brochures, and leaflets, with clarity and purpose relevant to scientific and professional contexts.
7. Define a research problem, formulate research questions or hypotheses, and conduct focused literature searches.
8. Apply ethical principles in scientific communication, recognize plagiarism risks, and correctly attribute sources and intellectual contributions.
9. Plan, design, and deliver compelling oral presentations, structuring content for clarity, managing nerves, and engaging audiences.
10. Design scientific visuals (figures, tables, infographics, posters) using tools like PowerPoint

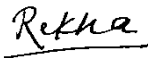
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and LaTeX Beamer, ensuring clarity and data integrity.

11. Use nonverbal communication effectively and respond to audience questions confidently during Q&A sessions.

BTH-87T-755 Food and Dairy Technology

Semester	Code of the Course	Title of the Course/Paper			NHEQ F Level	Credits
VII	BTH-87T-755	FOOD AND DAIRY TECHNOLOGY			8	4
Level of Course	Type of the Course	Credit Distribution			Offered to NC Student	Course Delivery Method
		Theory	Practical	Total		
Advanced	Major	4	2	6	NO	60 lectures with diagrammatic and informative assessments during lecture hours
List of Programme Codes in which Offered as Minor Discipline		-----				
Prerequisites		Knowledge of the Intermediate level				
Course Objectives		➤ To provide comprehensive knowledge of food processing, emphasizing the nutritive value and perishability of food products. ➤ To impart understanding of key unit operations and principles in dairy processing and dairy engineering. ➤ To explore the role of microorganisms in food production, processing, and preservation, including their use in fermented foods and beverages. ➤ To apply enzymatic and technological methods for enhancing food quality and shelf-life through post-harvest preservation. ➤ To explain milk composition, microbial quality control, and the impact of processing techniques like pasteurization and homogenization on dairy products. ➤ To develop practical knowledge of milk processing operations, plant sanitation, refrigeration, and equipment used in dairy technology.				

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Detailed Syllabus
BTH-87T-755 FOOD AND DAIRY TECHNOLOGY

Unit-I

Microbial role in food process: Production of single-cell protein and new protein foods: Mushroom, food yeast, and algal protein. Fermentation as a method for food preparation and preservation. Food additives: colours, flavours, and vitamins. Organisms and their use for the Production of Fermented Food and Beverages: Pickles, alcoholic beverages, cheese, sauerkraut, idli, and vinegar.

15 Lectures

Unit-II

Deoxygenating and desugaring by glucose oxidase, Beer Mashing & Chill Proofing, Cheese making by protease, Enzyme applications in food processing, Classification of Fruit Juice. Post-Harvest Technology and the process of food Preservation.

15 Lectures

Unit-III

Fundamentals of Dairy Science: Milk: Definition and Composition, Chemical & Functional Properties of milk components: Physicochemical properties of milk protein, aggregation of Casein micelles, factors affecting milk composition, milk secretion and lactation
Microorganism: Importance in dairy science and technology, Microbial Spoilage of Milk, Hydrolytic Rancidity of milk and milk products, Autoxidation of milk fats and effects on milk quality

15 Lectures

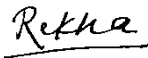
Unit-IV

Milk processing operation: Pasteurization: Homogenization, Sterilization. Effects of processing on milk components & their functional properties. Types of milk, Adulteration of milk.
Dairy Engineering: Cleaning & Sanitation: Cleaning agents, CIP & COP. Sanitary milk pump & fittings, Boiler: Properties of steam, steam generation, types of boilers. Refrigeration: Vapor compression refrigeration cycle, common refrigerants, and properties of good refrigerants. Dairy Plant layout, Types of cold storage.

15 Lectures

BTH-87P-756 FOOD AND DAIRY TECHNOLOGY–PRACTICAL

1. Parameters for quality control analysis of milk
2. Production of Sauerkraut
3. Determination of physical characteristics of milk
4. Determination of the chemical properties of milk
5. Determination of the biological characteristics of milk (MBRT test)
6. To detect the number of bacteria in milk by SPC
7. Various methods and tests for adulteration of milk
8. Removal of fat from milk
9. Isolation of proteins from milk

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10. Litmus milk test
11. Production of wine from grape juice
12. Educational tour to the Dairy
13. To study lipid or fat hydrolysis by microbes

Suggested Books & Text

1. Satyanarayana, U. Biotechnology. Elsevier
2. Patel, A.H. Industrial Microbiology. Macmillan India.
3. De, Sukumar. Outlines of Dairy Technology. Oxford University Press, 2014.
4. Singh, R.P. Microbiology. Kalyani Publishers.
5. Singh, B.D. Biotechnology: Expanding Horizons. Kalyani Publisher.
6. Frazier, William C., and Dennis C. Westhoff. Food Microbiology. McGraw-Hill Education, 2014.
7. Jay, James M., Martin J. Loessner, and David A. Golden. Modern Food Microbiology. Springer, 2005.
8. Banwart, George J. Basic Food Microbiology. Springer, 1989.
9. Ranganna, S. Handbook of Analysis and Quality Control for Fruit and Vegetable Products. Tata McGraw Hill, 1986.
10. Webb, B.H., A.H. Johnson, and J.A. Alford. Fundamentals of Dairy Chemistry. Springer, 1974.
11. De, Sukumar. Outlines of Dairy Technology. Oxford University Press, 2014.
12. Walstra, Pieter, Jan T.M. Wouters, and Tom J. Geurts. Dairy Science and Technology. 2nd ed., CRC Press, 2006.
13. Hui, Y.H., editor. Handbook of Food Science, Technology, and Engineering. CRC Press, 2006.
14. Fox, Patrick F., and Paul L.H. McSweeney. Dairy Chemistry and Biochemistry. Springer, 1998.
15. Joshi, V.K., and A. Pandey. Biotechnology: Food Fermentation (Microbiology, Biochemistry and Technology). Vols. I & II. Educational Publishers & Distributors, 1999.

Course Learning Outcomes:

By the end of the course, students will be able to:

1. Understand the role of microorganisms in producing, processing, and preserving various foods, including fermented products and single-cell proteins.
2. Apply enzymatic and technological methods for food processing and post-harvest preservation to enhance food quality and shelf life.
3. Explain fundamentals of dairy science, including milk composition, properties, and microbial quality control during dairy processing.
4. Demonstrate knowledge of milk processing operations and dairy engineering, including pasteurization, homogenization, sterilization, sanitation, refrigeration, and plant layout.
5. Develop value-added and functional food and dairy products using advanced processing techniques while maintaining nutritional quality and safety.

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